

Off-Diagonal Commonality of a Shorter Even Cycle and a Longer Odd Cycle

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Abstract

Let $m \geq 2$ and $k \geq m$, so that the even cycle C_{2m} is shorter than the odd cycle C_{2k+1} . We determine the complete off-diagonal commonality region of the ordered pair (C_{2m}, C_{2k+1}) . Let $a_{m,k}^* \in (1/2, 1)$ be the unique solution of

$$\frac{(a_{m,k}^*)^{2m-1}(2m + (2k + 1 - 2m)a_{m,k}^*)}{2k + 1} = 2^{1-2m}.$$

We prove that the commonality region is exactly $(0, a_{m,k}^*]$. The upper bound is witnessed by the balanced disjoint union of two cliques. For the lower bound, cycle densities are expressed as traces of graphon operators. Rank-one eigenvalue majorisation separates the Perron eigenvalue from the rest of the spectrum, while a Hilbert–Schmidt budget shows that at most one negative eigenvalue can contribute with the wrong sign. That eigenvalue is paid for by the Perron eigenvalue, leaving the elementary inequality $x^{2m} + (1 - x)^{2m} \geq 2^{1-2m}$.

1 Definitions and main theorem

A *graphon* is a symmetric measurable function $W: [0, 1]^2 \rightarrow [0, 1]$. If H is a finite graph, its homomorphism density in W is

$$t(H, W) := \int_{[0,1]^{V(H)}} \prod_{uv \in E(H)} W(x_u, x_v) \prod_{u \in V(H)} dx_u.$$

We write $e(H) = |E(H)|$. See Lovász [3] for background on graphons and homomorphism densities.

Following Behague, Morrison, and Noel [1], let $a, b \in (0, 1)$ with $a + b = 1$. An ordered pair (H_1, H_2) is (a, b) -*common* if

$$\frac{t(H_1, W)}{e(H_1)a^{e(H_1)-1}} + \frac{t(H_2, 1 - W)}{e(H_2)b^{e(H_2)-1}} \geq \frac{a}{e(H_1)} + \frac{b}{e(H_2)} \quad (1)$$

for every graphon W . Define

$$\pi(H_1, H_2) := \{a \in (0, 1) : (H_1, H_2) \text{ is } (a, 1 - a)\text{-common}\}.$$

Theorem 1.1 (Shorter even and longer odd cycles). *Let $m \geq 2$ and $k \geq m$, and put*

$$n := 2m, \quad \ell := 2k + 1.$$

Thus $n < \ell$. There is a unique $a_{n,\ell}^ \in (1/2, 1)$ satisfying*

$$\frac{(a_{n,\ell}^*)^{n-1}(n + (\ell - n)a_{n,\ell}^*)}{\ell} = 2^{1-n}. \quad (2)$$

Moreover,

$$\pi(C_n, C_\ell) = (0, a_{n,\ell}^*]. \quad (3)$$

Writing $a_{m,k}^* := a_{2m,2k+1}^*$, the critical equation in terms of the indices in the title is

$$\frac{(a_{m,k}^*)^{2m-1}(2m + (2k + 1 - 2m)a_{m,k}^*)}{2k + 1} = 2^{1-2m}.$$

Some numerical values of the critical point are displayed in Table 1. The columns give the even length n and the rows give the odd length ℓ . An entry is replaced by 0 when $n \geq \ell$, which lies outside the shorter-even regime of Theorem 1.1.

$\ell \backslash n$	4	6	8	10	12	14
5	0.517213...	0	0	0	0	0
7	0.538147...	0.507351...	0	0	0	0
9	0.550283...	0.517832...	0.504065...	0	0	0
11	0.558163...	0.524929...	0.510344...	0.502577...	0	0
13	0.563680...	0.530044...	0.514968...	0.506759...	0.501779...	0
15	0.567753...	0.533903...	0.518513...	0.510006...	0.504764...	0.501302...

Table 1: The critical values $a_{n,\ell}^*$ for shorter even cycles C_n and longer odd cycles C_ℓ .

The proof of Theorem 1.1 is self-contained apart from standard finite-dimensional spectral facts.

2 Reduction to finite-dimensional operators

We first show that it is enough to prove the required inequalities for step graphons.

Lemma 2.1 (Step-graphon reduction). *Fix integers $r, s \geq 3$ and a continuous function $\Phi: \mathbb{R}^2 \rightarrow \mathbb{R}$. Suppose that*

$$\Phi(t(C_r, W), t(C_s, 1 - W)) \geq 0$$

for every symmetric step graphon W . Then the same inequality holds for every graphon W .

Proof. For $j \geq 1$, partition $[0, 1]$ into its 2^j dyadic intervals of equal length, and let $\mathcal{A}_j$ be the finite sigma-algebra generated by this partition. Set

$$W_j := \mathbb{E}[W \mid \mathcal{A}_j \otimes \mathcal{A}_j].$$

Conditional expectation preserves symmetry and the pointwise bounds $0 \leq W \leq 1$, so W_j is a symmetric step graphon.

The dyadic rectangle step functions form nested subspaces whose union is dense in $L^2([0, 1]^2)$. Since conditional expectation is the orthogonal projection onto the corresponding subspace,

$$\|W_j - W\|_2 \rightarrow 0.$$

The underlying space has measure one, so also $\|W_j - W\|_1 \rightarrow 0$.

For every fixed $q \geq 3$, the telescoping identity for products and the bounds $0 \leq W_j, W \leq 1$ give

$$|t(C_q, W_j) - t(C_q, W)| \leq q\|W_j - W\|_1. \quad (4)$$

Indeed, after writing $x_{q+1} = x_1$, expand the difference of the two products into q terms, with the i th term containing $(W_j - W)(x_i, x_{i+1})$ and all other factors bounded in absolute value by one. Integration of each term gives $\|W_j - W\|_1$. The same estimate applies to $1 - W_j$ and $1 - W$. Both cycle densities therefore converge, and continuity of Φ proves the lemma. $\square$

Let K be a symmetric step kernel, constant on every rectangle $B_i \times B_j$ of a finite measurable partition

$$[0, 1] = B_1 \sqcup \cdots \sqcup B_N, \quad |B_i| > 0.$$

Let T_K be its integral operator on $L^2([0, 1])$,

$$(T_K f)(x) = \int_0^1 K(x, y) f(y) dy.$$

Put

$$E := \text{span}\{\mathbf{1}_{B_1}, \dots, \mathbf{1}_{B_N}\}.$$

The operator T_K maps $L^2([0, 1])$ into E and vanishes on $E^\perp$. In the orthonormal basis

$$e_i := \frac{\mathbf{1}_{B_i}}{\sqrt{|B_i|}},$$

the restriction $T_K|_E$ is represented by the real symmetric matrix

$$(K_{ij} \sqrt{|B_i| |B_j|})_{i,j=1}^N,$$

where K_{ij} is the value of K on $B_i \times B_j$.

Lemma 2.2 (Cycle densities are traces). *For every integer $q \geq 3$,*

$$t(C_q, K) = \text{Tr}(T_K^q), \tag{5}$$

where the trace is taken on E .

Proof. Expanding the matrix trace in the basis $(e_i)_{i=1}^N$ gives

$$\text{Tr}(T_K^q) = \sum_{i_1, \dots, i_q=1}^N |B_{i_1}| \cdots |B_{i_q}| K_{i_1 i_2} K_{i_2 i_3} \cdots K_{i_q i_1}.$$

This is precisely the step-function expansion of $t(C_q, K)$. □

Let $J := T_1$. Since the measure of $[0, 1]$ is one,

$$Jf = \langle \mathbf{1}, f \rangle \mathbf{1},$$

so J is the rank-one orthogonal projection onto the constant functions. If $U = 1 - W$ and $T = T_U$, then

$$T_W = J - T. \tag{6}$$

3 Rank-one majorisation and spectral bounds

For a vector $x \in \mathbb{R}^N$, write $x^\downarrow$ for its coordinates rearranged in nonincreasing order. Recall that x is *majorised* by y , written $x \prec y$, if

$$\sum_{i=1}^j x_i^\downarrow \leq \sum_{i=1}^j y_i^\downarrow \quad (1 \leq j < N), \quad \sum_{i=1}^N x_i = \sum_{i=1}^N y_i.$$

Karamata's inequality [4, Chapter 4] states that, whenever $x \prec y$ and $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ is convex,

$$\sum_{i=1}^N \varphi(x_i) \leq \sum_{i=1}^N \varphi(y_i). \tag{7}$$

We need the following rank-one case of Lidskii's additive majorisation theorem. A proof is included for completeness.

Lemma 3.1 (Rank-one lower majorisation). *Let A be a real symmetric $N \times N$ matrix with eigenvalues*

$$\alpha_1 \geq \alpha_2 \geq \cdots \geq \alpha_N,$$

and let P be a rank-one orthogonal projection. If

$$\mu_1 \geq \mu_2 \geq \cdots \geq \mu_N$$

are the eigenvalues of $A + P$, then

$$(\alpha_1, \dots, \alpha_{N-1}, \alpha_N + 1) \prec (\mu_1, \dots, \mu_N). \quad (8)$$

Proof. Eigenvalues under a positive rank-one perturbation interlace:

$$\mu_i \geq \alpha_i \geq \mu_{i+1} \quad (1 \leq i < N), \quad \mu_N \geq \alpha_N. \quad (9)$$

These inequalities follow from the Courant–Fischer min–max principle; see, for example, [2, Chapter III].

Put

$$v := (\alpha_1, \dots, \alpha_{N-1}, \alpha_N + 1).$$

Fix $1 \leq j < N$. The sum of the j largest coordinates of v is the maximum of $\sum_{i \in S} v_i$ over all j -element subsets $S \subseteq \{1, \dots, N\}$.

If $N \notin S$, then

$$\sum_{i \in S} v_i \leq \sum_{i=1}^j \alpha_i \leq \sum_{i=1}^j \mu_i.$$

If $N \in S$, then

$$\sum_{i \in S} v_i \leq 1 + \alpha_N + \sum_{i=1}^{j-1} \alpha_i.$$

On the other hand, $\text{Tr}(A + P) = \text{Tr}(A) + 1$, and $\mu_{i+1} \leq \alpha_i$ for $j \leq i < N$. Hence

$$\begin{aligned} \sum_{i=1}^j \mu_i &= \text{Tr}(A + P) - \sum_{i=j+1}^N \mu_i \\ &\geq 1 + \sum_{i=1}^N \alpha_i - \sum_{i=j}^{N-1} \alpha_i \\ &= 1 + \alpha_N + \sum_{i=1}^{j-1} \alpha_i. \end{aligned}$$

Thus every partial-sum inequality in the definition of majorisation holds. Equality of the total sums again follows from $\text{Tr}(A + P) = \text{Tr}(A) + 1$. $\square$

Corollary 3.2 (Even trace after subtracting a rank-one projection). *Let T be a real symmetric $N \times N$ matrix with eigenvalues*

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N,$$

and let P be a rank-one orthogonal projection. If $q \geq 2$ is even, then

$$\text{Tr}(P - T)^q \geq (1 - \lambda_1)^q + \sum_{i=2}^N |\lambda_i|^q. \quad (10)$$

Proof. Apply Lemma 3.1 to $A = -T$. The eigenvalues of $-T$ in nonincreasing order are

$$-\lambda_N \geq \dots \geq -\lambda_1.$$

Consequently,

$$(-\lambda_N, \dots, -\lambda_2, 1 - \lambda_1) \prec \lambda(P - T),$$

where $\lambda(P - T)$ denotes the eigenvalue vector of $P - T$. Apply Karamata's inequality to the convex function $x \mapsto x^q$. Since q is even, $(-\lambda_i)^q = |\lambda_i|^q$, and the result follows. $\square$

The next estimates record all the consequences of $0 \leq U \leq 1$ that will be used later. Notice that T_U need not be positive semidefinite.

Lemma 3.3 (Perron domination and the spectral budget). *Let U be a symmetric step graphon and let $T = T_U|_E$ be its matrix on an associated step-function space. If*

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N$$

are the eigenvalues of T , then

$$0 \leq \lambda_1 \leq 1, \tag{11}$$

$$|\lambda_i| \leq \lambda_1 \quad (1 \leq i \leq N), \tag{12}$$

$$\sum_{i=2}^N \lambda_i^2 \leq \lambda_1(1 - \lambda_1) \leq \frac{1}{4}. \tag{13}$$

Proof. If $f \in E$ is a real unit vector, then $|f| \in E$ is also a unit vector. Pointwise nonnegativity of U gives

$$|\langle f, Tf \rangle| \leq \langle |f|, T|f| \rangle \leq \lambda_1, \tag{14}$$

where the second inequality is the Rayleigh principle. Applying this to a unit eigenvector of λ_i proves $|\lambda_i| \leq \lambda_1$. Also, $\lambda_1 \geq 0$ by testing the nonnegative vector $\mathbf{1}$. Since $U \leq 1$,

$$\langle |f|, T|f| \rangle \leq \left(\int_0^1 |f| \right)^2 \leq \|f\|_2^2 = 1,$$

which gives $\lambda_1 \leq 1$.

Finally, put

$$p := \int_{[0,1]^2} U(x, y) dx dy = \langle \mathbf{1}, T\mathbf{1} \rangle.$$

The Rayleigh principle gives $p \leq \lambda_1$, while $U^2 \leq U$ gives

$$\sum_{i=1}^N \lambda_i^2 = \text{Tr}(T^2) = \int_{[0,1]^2} U(x, y)^2 dx dy \leq p.$$

Therefore

$$\sum_{i=2}^N \lambda_i^2 \leq p - \lambda_1^2 \leq \lambda_1 - \lambda_1^2 \leq \frac{1}{4}.$$

$\square$

4 The scalar critical point when the odd cycle is longer

Throughout this section, let

$$n \geq 4 \text{ be even,} \quad \ell > n \text{ be odd,} \quad d := \ell - n.$$

Thus d is a positive odd integer. For $a \in (0, 1)$, set $b := 1 - a$ and define

$$\kappa_{n,\ell}(a) := \frac{na^{n-1}}{\ell b^{\ell-1}}, \quad (15)$$

$$\rho_{n,\ell}(a) := a^n + \frac{n}{\ell}a^{n-1}b = \frac{a^{n-1}(n + da)}{\ell}. \quad (16)$$

Multiplying the commonality inequality by na^{n-1} shows that (C_n, C_ℓ) is (a, b) -common exactly when

$$t(C_n, W) + \kappa_{n,\ell}(a)t(C_\ell, 1 - W) \geq \rho_{n,\ell}(a) \quad (17)$$

for every graphon W .

For $\kappa > 0$, define

$$f_\kappa(x) := (1 - x)^n + \kappa x^\ell, \quad 0 \leq x \leq 1. \quad (18)$$

If $\kappa = \kappa_{n,\ell}(a)$, then

$$\begin{aligned} f'_\kappa(x) &= -n(1 - x)^{n-1} + \kappa \ell x^{\ell-1}, \\ f''_\kappa(x) &= n(n - 1)(1 - x)^{n-2} + \kappa \ell(\ell - 1)x^{\ell-2} > 0. \end{aligned}$$

Moreover,

$$f'_\kappa(b) = -na^{n-1} + \kappa \ell b^{\ell-1} = 0.$$

Thus f_κ is strictly convex and has its unique minimum at b , with

$$f_\kappa(b) = a^n + \kappa b^\ell = \rho_{n,\ell}(a). \quad (19)$$

Lemma 4.1 (Existence and uniqueness of the critical point). *There is a unique $a_{n,\ell}^* \in (1/2, 1)$ such that*

$$\rho_{n,\ell}(a_{n,\ell}^*) = 2^{1-n}. \quad (20)$$

Proof. Differentiating (16) gives

$$\rho'_{n,\ell}(a) = \frac{n}{\ell}a^{n-2}(n - 1 + da) > 0.$$

Hence $\rho_{n,\ell}$ is strictly increasing. At the endpoints,

$$\rho_{n,\ell}\left(\frac{1}{2}\right) = 2^{1-n}\frac{n + \ell}{2\ell} < 2^{1-n}, \quad \rho_{n,\ell}(1) = 1 > 2^{1-n}.$$

The intermediate value theorem proves existence, and strict monotonicity proves uniqueness. $\square$

The following endpoint estimates replace all length-specific numerical bounds in the spectral argument.

Lemma 4.2 (Critical coefficient bounds). *Put*

$$a^* := a_{n,\ell}^*, \quad b^* := 1 - a^*, \quad \kappa^* := \kappa_{n,\ell}(a^*).$$

Then

$$\kappa^*(b^*)^d < 1, \quad b^* > \frac{5}{14}. \quad (21)$$

Consequently, for every $0 < a \leq a^$, with $b = 1 - a$ and $\kappa = \kappa_{n,\ell}(a)$,*

$$\kappa b^d < 1, \quad b > \frac{5}{14}. \quad (22)$$

Proof. Write

$$a^* = \frac{1+\delta}{2}, \quad b^* = \frac{1-\delta}{2}, \quad \delta \in (0, 1).$$

Since $\ell = n + d$, the critical equation (20) is equivalent to

$$(1+\delta)^{n-1}(\ell + n + d\delta) = 2\ell. \quad (23)$$

It follows that

$$(1+\delta)^{n-1} < \frac{2\ell}{\ell+n} = 1 + \frac{d}{\ell+n}.$$

Bernoulli's inequality therefore yields

$$0 < \delta < \frac{d}{(n-1)(\ell+n)}. \quad (24)$$

A second application of Bernoulli's inequality gives

$$(1-\delta)^{n-1} \geq 1 - (n-1)\delta > 1 - \frac{d}{\ell+n} = \frac{2n}{\ell+n}. \quad (25)$$

Combining these two estimates gives

$$\begin{aligned} \kappa^*(b^*)^d &= \frac{n}{\ell} \left(\frac{1+\delta}{1-\delta} \right)^{n-1} \\ &< \frac{n}{\ell} \frac{2\ell/(\ell+n)}{2n/(\ell+n)} = 1. \end{aligned}$$

Equation (23) also implies $(1+\delta)^{n-1} < 2$. Since $n-1 \geq 3$ and $1+\delta > 1$,

$$(1+\delta)^3 \leq (1+\delta)^{n-1} < 2.$$

If $\delta \geq 2/7$, then

$$(1+\delta)^3 \geq \left(\frac{9}{7}\right)^3 = \frac{729}{343} > 2,$$

a contradiction. Thus $\delta < 2/7$, and hence

$$b^* = \frac{1-\delta}{2} > \frac{5}{14}.$$

Finally,

$$\kappa_{n,\ell}(a)(1-a)^d = \frac{n}{\ell} \left(\frac{a}{1-a} \right)^{n-1}$$

is strictly increasing on $(0, 1)$. Therefore $\kappa b^d \leq \kappa^*(b^*)^d < 1$ whenever $a \leq a^*$. Moreover, $b = 1-a \geq 1-a^* = b^* > 5/14$. $\square$

5 The exact region when the odd cycle is longer

We retain the assumptions and notation of Section 4.

Proposition 5.1 (Spectral reduction). *Let W be a symmetric step graphon, put $U = 1 - W$, and let $T = T_U|_E$ on an associated step-function space. Order the eigenvalues of T as*

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N.$$

For every $\kappa > 0$,

$$t(C_n, W) + \kappa t(C_\ell, U) \geq f_\kappa(\lambda_1) + \sum_{i=2}^N \lambda_i^n (1 + \kappa \lambda_i^d). \quad (26)$$

Proof. Let $P := J|_E$. Since $\mathbf{1} \in E$ and $\|\mathbf{1}\|_2 = 1$, P is a rank-one orthogonal projection. By (6), $T_W|_E = P - T$. Lemma 2.2 and Corollary 3.2 give

$$t(C_n, W) = \text{Tr}(P - T)^n \geq (1 - \lambda_1)^n + \sum_{i=2}^N |\lambda_i|^n.$$

Because n is even, $|\lambda_i|^n = \lambda_i^n$. Also,

$$t(C_\ell, U) = \text{Tr}(T^\ell) = \sum_{i=1}^N \lambda_i^\ell.$$

Since $\ell = n + d$, adding the last two displays gives (26). $\square$

Proposition 5.2 (Lower bound up to the critical point). *If $0 < a \leq a_{n,\ell}^*$, then every graphon W satisfies*

$$t(C_n, W) + \kappa_{n,\ell}(a)t(C_\ell, 1 - W) \geq \rho_{n,\ell}(a). \quad (27)$$

Proof. Put $b = 1 - a$ and $\kappa = \kappa_{n,\ell}(a)$. By Lemma 2.1, it is enough to consider a symmetric step graphon W . Let $U = 1 - W$, let $T = T_U|_E$, and write the eigenvalues of T as

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N.$$

Proposition 5.1 gives

$$t(C_n, W) + \kappa t(C_\ell, U) \geq f_\kappa(\lambda_1) + \sum_{i=2}^N \lambda_i^n (1 + \kappa \lambda_i^d). \quad (28)$$

Call an eigenvalue λ_i , $i \geq 2$, *dangerous* if

$$1 + \kappa \lambda_i^d < 0. \quad (29)$$

Since d is odd and $\kappa > 0$, a dangerous eigenvalue has the form $\lambda_i = -\beta$ with $\beta > 0$. Its defining inequality gives

$$\kappa \beta^d > 1 > \kappa b^d. \quad (30)$$

Since $x \mapsto x^d$ is strictly increasing on $[0, \infty)$, it follows that

$$\beta > b > \frac{5}{14}. \quad (31)$$

Every nondangerous summand in (28) is nonnegative because n is even.

There is at most one dangerous index. Indeed, if two distinct dangerous indices existed, with magnitudes β and γ , then Lemma 3.3 and (31) would give

$$\frac{1}{4} \geq \sum_{i=2}^N \lambda_i^2 \geq \beta^2 + \gamma^2 > 2 \left(\frac{5}{14} \right)^2 = \frac{25}{98} > \frac{1}{4}.$$

If there is no dangerous eigenvalue, then (28) and (19) immediately give

$$t(C_n, W) + \kappa t(C_\ell, U) \geq f_\kappa(\lambda_1) \geq f_\kappa(b) = \rho_{n,\ell}(a).$$

Suppose instead that the unique dangerous eigenvalue is $-\beta$. Perron domination (12) gives

$$\beta \leq \lambda_1 \leq 1. \quad (32)$$

Moreover, (31) gives

$$\beta > b. \quad (33)$$

The strictly convex function f_κ is increasing on $[b, 1]$. Therefore (32) and (33) imply

$$f_\kappa(\lambda_1) \geq f_\kappa(\beta).$$

All other tail terms in (28) are nonnegative, so the dangerous term can be combined with f_κ as follows:

$$\begin{aligned} t(C_n, W) + \kappa t(C_\ell, U) &\geq f_\kappa(\beta) + \beta^n(1 - \kappa\beta^d) \\ &= (1 - \beta)^n + \kappa\beta^\ell + \beta^n - \kappa\beta^{n+d} \\ &= (1 - \beta)^n + \beta^n \\ &\geq 2^{1-n}. \end{aligned}$$

The last inequality follows from convexity of $x \mapsto x^n$ applied to β and $1 - \beta$. Finally, monotonicity of $\rho_{n,\ell}$ and $a \leq a_{n,\ell}^*$ imply

$$\rho_{n,\ell}(a) \leq \rho_{n,\ell}(a_{n,\ell}^*) = 2^{1-n}.$$

This proves (27) in the remaining case. $\square$

We now give the matching obstruction. Let W_{cl} be the balanced disjoint union of two cliques:

$$W_{\text{cl}}(x, y) := \begin{cases} 1, & x, y \in [0, 1/2] \text{ or } x, y \in (1/2, 1], \\ 0, & \text{otherwise.} \end{cases}$$

Every homomorphism from the connected graph C_n into W_{cl} maps all vertices into one of the two halves. Thus

$$t(C_n, W_{\text{cl}}) = 2 \left(\frac{1}{2} \right)^n = 2^{1-n}. \quad (34)$$

The complement $1 - W_{\text{cl}}$ is the balanced complete bipartite graphon. Since ℓ is odd,

$$t(C_\ell, 1 - W_{\text{cl}}) = 0. \quad (35)$$

If $a > a_{n,\ell}^*$, strict monotonicity of $\rho_{n,\ell}$ gives

$$\rho_{n,\ell}(a) > 2^{1-n}.$$

Equations (34) and (35) therefore show that W_{cl} violates (17). Together with Proposition 5.2, this proves

$$\pi(C_n, C_\ell) = (0, a_{n,\ell}^*]$$

whenever $n < \ell$.

Proof of Theorem 1.1. Lemma 4.1 gives the unique critical point. Proposition 5.2 proves commonality for every a up to and including that point, and the balanced two-clique obstruction proves failure for every larger a . $\square$

6 Remarks

The proof of Theorem 1.1 has been verified in the Lean 4 proof assistant [6], on top of the mathlib library [5]; the development accompanies this note. The formal statement covers both directions and the existence and uniqueness of $a_{n,\ell}^*$: the inequality holds for every graphon if and only if $a \leq a_{n,\ell}^*$. The formal proof differs from the one given here in four ways.

First, instead of the dyadic conditional expectations in Lemma 2.1, the formal proof approximates W directly by a step graphon in the L^1 norm; both cycle densities are Lipschitz in that norm, so the inequality transfers.

Second, the interlacing inequalities (9) are proved by an intersection argument instead of the Courant–Fischer principle: two spans of eigenvectors whose dimensions add up to more than N contain a common unit vector, and evaluating the quadratic forms of A and $A + P$ at that vector gives the inequality.

Third, the rearrangement in Lemma 3.1 is not obtained by sorting. Since the α_i are already ordered, rearranging $(\alpha_1, \dots, \alpha_{N-1}, \alpha_N + 1)$ only moves the entry $\alpha_N + 1$ to its correct position, and the formal proof writes this down explicitly, with the partial sums in closed form. Karamata’s inequality (7) is proved in a form that requires only one of the two sequences to be monotone, so no second sorting argument is needed.

Fourth, the derivatives of f_κ are not computed. The paper locates the minimum of f_κ at b through f'_κ and f''_κ ; the formal proof instead bounds $(1 - x)^n$ and κx^ℓ from below by their supporting lines at $x = b$, both instances of Bernoulli’s inequality, and the sum of the two lines is the constant $\rho_{n,\ell}(a)$ because $\kappa \ell b^{\ell-1} = n a^{n-1}$. The same supporting-line estimate, together with monotonicity of powers on the nonnegative reals, proves that f_κ is increasing on $[b, 1]$. Strict monotonicity of $\rho_{n,\ell}$ is proved directly from its explicit formula, without differentiating. Finally, the formal proof obtains $x^n + (1 - x)^n \geq 2^{1-n}$ by applying the supporting-line inequality at $1/2$ to both terms.

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